



Course : R-VIJETA (R-JP)

PHYSICS DPP

DAILY PRACTICE PROBLEM

REVISION DPP NO. # 05

Target : JEE(Advanced) 2023

DPP Syllabus :

SURFACE TENSION, FLUID MECHANICS, ELASTICITY, VISCOSITY & SHM

DPP No. : 05

Total Marks : 170

One or more than one may be correct option ('-1' Negative marking) Q.1 to 20;

Subjective questions ('-1' Negative marking) Q.21 to Q.40 ;

Comprehension ('-1' negative marking) Q.41 to Q.44

Single choice Objective ('-1' negative marking) Q.45 to Q.50

Max. Time : 160 min.

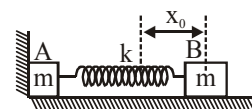
(3 Marks; 2 Min.) [60, 40]

(4 Marks; 5 Min.) [80, 100]

(3 Marks; 2 Min.) [12, 08]

(3 Marks; 2 Min.) [18, 12]

1. Two identical blocks A and B each of mass m are connected to each other by spring of spring constant k . Block B is initially shifted to a small distance x_0 from natural length of spring to the left and then released. Choose the correct statements for this problem, after the spring attains its natural length.



(A) Velocity of centre of mass of the system is $\frac{1}{2} \sqrt{\frac{k}{m}} x_0$.

(B) Maximum elongation in spring during the subsequent motion is $\frac{x_0}{\sqrt{2}}$.

(C) Maximum elongation in spring during the subsequent motion is x_0 .

(D) Maximum speed of block A during subsequent motion will be $\sqrt{\frac{k}{m}} x_0$.

2. When a capillary tube is dipped in a liquid, the liquid rises to a height h in the tube. The free liquid surface inside the tube is hemispherical in shape. The tube is now pushed down so that the height of the tube outside the liquid is less than h .

(A) The liquid will come out of the tube like in a small fountain.

(B) The liquid will come out of the tube slowly.

(C) The liquid will fill the tube but not come out of its upper end.

(D) The free liquid surface inside the tube will not be hemispherical.

3. A light rod of length ℓ is suspended from ceiling horizontally using two vertical wires of equal length tied to its ends. One of the wires is made of steel of cross-sectional area a_s and young's modulus Y_s . While the other wire is made of brass of cross-sectional area a_b and Young's modulus Y_b , then :

(A) For stress to be same, a mass can be hanged at distance $\frac{\ell a_s}{a_b + a_s}$ from brass wire end.

(B) For stress to be same, a mass can be hanged at distance $\frac{\ell a_s}{a_b + a_s}$ from steel wire end.

(C) For equal strain, the mass should be hanged at mid point of the rod.

(D) For equal strain, the mass should be at distance $\frac{\ell a_s Y_s}{a_s Y_s + a_b Y_b}$ from brass wire end.



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4. The velocity of liquid (v) in steady flow at a location through cylindrical pipe is given by $v = v_0 \left(1 - \frac{r^2}{R^2} \right)$, where r is the radial distance of that location from the axis of the pipe and R is the inner radius of pipe. If $R = 10$ cm, volume rate of flow through the pipe is $\pi/2 \times 10^{-2} \text{ m}^3 \text{ s}^{-1}$ and the coefficient of viscosity of the liquid is 0.75 N.s/m^2 :
- (A) Magnitude of viscous force per unit area at $r = 4$ cm is 6 N/m^2 .
 (B) Maximum velocity of flow of liquid is 1 m/s .
 (C) Viscous force per unit area increases with radial distance.
 (D) Viscous force per unit area decreases with radial distance.

5. A simple pendulum A and a homogeneous rod B hinged at its end are released from horizontal position as shown in figure.



- (A) The time period of swing of simple pendulum is greater than that of rod if $L = \ell$.
 (B) The time period of swing of simple pendulum is equal to that of rod if $L = \ell$.
 (C) The time period of swing of simple pendulum is greater than that of rod if $L = \frac{2\ell}{3}$.
 (D) The time period of swing of simple pendulum is equal to that of rod if $L = \frac{2\ell}{3}$.

6. Two spherical soap bubbles in vacuum are connected through a narrow tube. Radius of left bubble is R and that of other is slightly smaller than R . Air flows from right to left very slowly. At any instant r_1, A_1, V_1, n_1 are radius, surface area, volume and number of moles of gas in the left bubble and r_2, A_2, V_2, n_2 are same for right bubble. Assume that temperature remains constant :



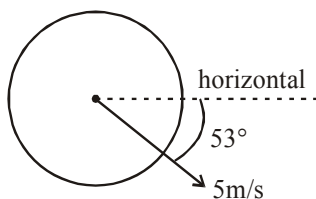
Suppose at any instant number of moles in left bubble is 4 times of number of moles in right bubble, then select correct statement(s) :

- (A) $r_2 = R\sqrt{\frac{2}{5}}$ (B) $r_1 = R\sqrt{\frac{8}{5}}$ (C) $A_1 + A_2 = \text{constant}$ (D) $V_1 + V_2 = \text{constant}$

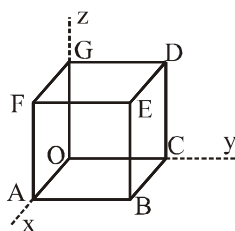
7. A metal wire of length L , area of cross-section A and Young's modulus Y is stretched by a variable force F such that F is always slightly greater than the elastic force of resistance in the wire. When the elongation of the wire is ℓ :

- (A) the work done by F is $\frac{YA^2}{L}$.
 (B) the work done by F is $\frac{YA\ell^2}{2L}$.
 (C) the elastic potential energy stored in the wire is $\frac{YA\ell^2}{2L}$.
 (D) heat is produced during the elongation.

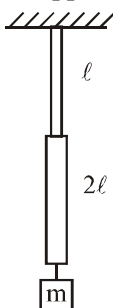
8. An external force 6 N is applied on a sphere of radius $R = 10$ cm of mass 1 kg and the sphere moves in a liquid with a constant velocity 5 m/s making 53° with the horizontal. The coefficient of viscosity of the liquid is $\frac{20}{6\pi}$ in S.I. units. (Take $g = 10 \text{ m/s}^2$). Choose the correct option(s) :



- (A) The viscous force on the body is 10 N.
 (B) The effective weight (weight – upthrust) of the body is 8 N.
 (C) The direction of the external applied force must be horizontal.
 (D) If the external force is suddenly removed the acceleration of the body just after the removal of the force will be 6 m/s^2 .
9. Cubical vessel is completely filled with water. Acceleration due to gravity is g . xy -plane is horizontal. Cube is given an acceleration $\vec{a} = +g \hat{i} - g \hat{j}$. An air bubble rises from corner C and a metal ball is released from corner G. Then :

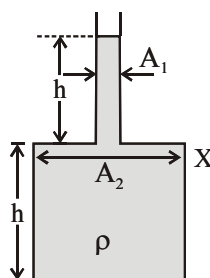


- (A) air bubble rises along the body diagonal CF. (B) air bubble rises along face diagonal CG.
 (C) metal ball moves along body diagonal GB. (D) metal ball moves along face diagonal GC.
10. A light wire of length ℓ and radius r is welded to another light wire of length 2ℓ and radius $2r$. The free end of 1st wire is fixed and the free end of 2nd wire supports a mass m . The Young's modulus of each wire is Y .

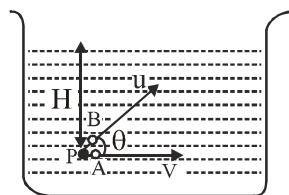


- (A) Extension in upper wire is $\frac{mg\ell}{\pi r^2 Y}$. (B) Extension in lower wire is $\frac{mg\ell}{2\pi r^2 Y}$.
 (C) Total elongation is $\frac{5}{2} \frac{mg\ell}{\pi r^2 Y}$. (D) Total elongation is $\frac{3}{2} \frac{mg\ell}{\pi r^2 Y}$.

11. In a certain gravity free space, the piston of an injection is being pushed so that the water jet comes out with a constant speed v . The area of the piston is much greater than the orifice of the injection.
- (A) The force required to be applied on the piston is proportional to v^2 .
- (B) The power developed by the force pushing the piston is proportional to v^3 .
- (C) The time for emptying the injection is proportional to v^{-1} .
- (D) The total work done in emptying the injection is proportional to v^2 .
12. The vessel shown in the figure has two sections of areas of cross-section A_1 and A_2 . A liquid of density ρ fills both the sections, up to a height h in each. Neglect atmospheric pressure.

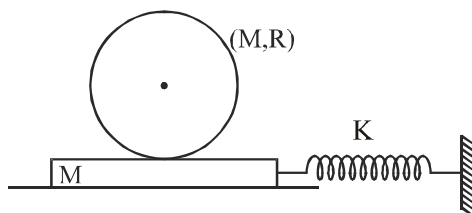


- (A) the pressure at the base of the vessel is $2h\rho g$.
- (B) the force exerted by the liquid on the base of the vessel is $2h\rho g A_2$.
- (C) the weight of the liquid is $< 2h\rho g A_2$.
- (D) the walls of the vessel at the level X exert a downward force $h\rho g(A_2 - A_1)$ on the liquid.
13. Two small balls A and B are projected simultaneously from point P inside liquid as shown. A is projected horizontally with speed V and B at an angle θ to the horizontal with speed u . Liquid has density ρ , ball A has density $\frac{\rho}{2}$ and ball B has density ρ . $H = 10$ m, $\theta = 45^\circ$, $g = 10$ m/s². If the balls collide each other at the surface of the liquid, then:[neglect viscous forces] :



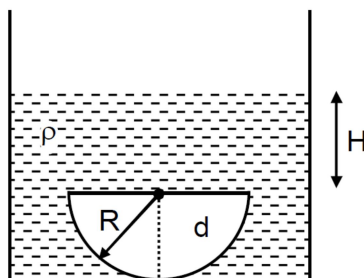
- (A) $u = 10\sqrt{2}$ m/s (B) $u = 10$ m/s (C) $u = 5$ m/s (D) $V = 5\sqrt{2}$ m/s
14. A small ball bearing is released at the top of a long vertical column of glycerine of height $2h$. The ball bearing falls through a height h in a time t_1 and then the remaining height with the terminal velocity in time t_2 . Let W_1 and W_2 be the work done against viscous drag over these height. Therefore:
- (A) $t_1 < t_2$ (B) $t_1 > t_2$ (C) $W_1 = W_2$ (D) $W_1 < W_2$

15. A solid cylinder of mass M and radius R lies on a plank of mass M as shown. The surface between plank and ground is smooth, and between cylinder and plank is rough. Assuming no slipping between cylinder and plank, and velocity amplitude of oscillation of plank is V .

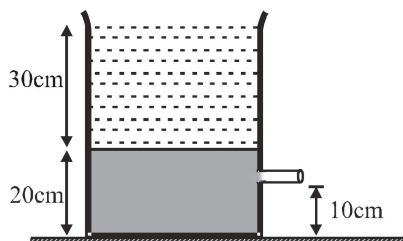


- (A) Time period of oscillation of the set up when displaced from equilibrium is $4\pi\sqrt{\frac{M}{3K}}$.
- (B) Friction between plank and cylinder when acceleration is maximum is $\frac{V}{2}\sqrt{\frac{MK}{3}}$.
- (C) Time period of oscillation of the set up when displaced from equilibrium is $2\pi\sqrt{\frac{M}{3K}}$.
- (D) Friction between plank and cylinder when acceleration is maximum is $\frac{V}{4}\sqrt{\frac{MK}{3}}$.
16. A particle constrained to move along x-axis is given a velocity 'u' at origin along the positive x-axis. The acceleration 'a' of the particle varies as $a = -bx$, where b is a positive constant and x is the x coordinate of the position of the particle. Then select the correct alternative(s) :
- (A) The maximum displacement of the particle from the starting point is $\frac{u}{\sqrt{b}}$.
- (B) The particle will oscillate about the origin.
- (C) Velocity is maximum at the origin.
- (D) Given data is insufficient to determine the exact motion of the particle.
17. A 20 gm particle is subjected to two simple harmonic motions $x_1 = 2 \sin 10t$, $x_2 = 4 \sin (10t + \frac{\pi}{3})$, where x_1 & x_2 are in metre & t is in sec :
- (A) The displacement of the particle at $t = 0$ will be $2\sqrt{3}$ m.
- (B) Maximum speed of the particle will be $20\sqrt{7}$ m/s.
- (C) Magnitude of maximum acceleration of the particle will be $200\sqrt{7}$ m/s².
- (D) Energy of the resultant simple harmonic motion will be 28 J.

18. A solid glass hemisphere of density d and radius R lies (with curved surface of hemisphere below the flat surface) at the bottom of a tank filled with water of density ρ such that the flat surface of hemisphere is H depth below the liquid surface. Combined weight of water + tank is W_1 and that of hemisphere is W_2 . Then choose the incorrect options :

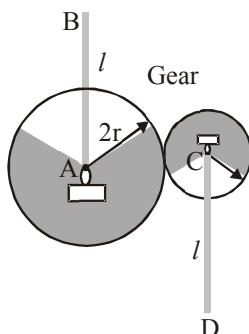


- (A) Force exerted by the liquid on the flat surface of hemisphere is independent of H and d but depends on R and ρ .
- (B) Force exerted by the liquid on the curved surface of hemisphere is independent of H and d but depends on R and ρ .
- (C) Force exerted by the liquid on the hemisphere is independent of H and d but depends on R and ρ .
- (D) Combined weight of water + tank + hemisphere with hemisphere inside water, taken by a weighing machine is equal to $W_1 + W_2$.
19. A cylindrical vessel of a very large cross-sectional area 1000 cm^2 is containing two immiscible liquids of density $\rho_1 = 600 \text{ kg/m}^3$ & $\rho_2 = 1200 \text{ kg/m}^3$ as shown in figure. A small hole having cross-sectional area 5 cm^2 is made in right side vertical wall. (Neglect the mass of container) For the given situation, mark the correct statement (s). (Take $g = 10 \text{ m/s}^2$)

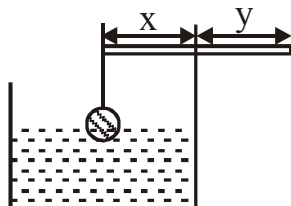


- (A) If the surface is smooth then a right ward force of magnitude 4 N is to be applied on vessel to maintain its equilibrium.
- (B) If the surface is smooth then a right ward force of magnitude 3 N is to be applied on vessel to maintain its equilibrium.
- (C) If the surface is rough ($\mu = 0.04$), then minimum force (horizontal) needed to be applied on the vessel to maintain equilibrium is zero.
- (D) If the surface is rough ($\mu = 0.04$), then the maximum force (horizontal) that can be applied on the vessel while maintaining equilibrium is 19.8 N.

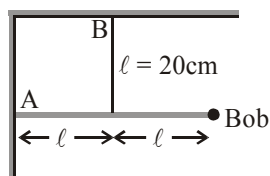
20. Assume that the potential energy of a di-atomic molecule is given by a two term expression, composed of a binding term $\frac{b}{r^3}$ & short range repulsion term $\frac{a}{r^5}$ where 'r' is distance between atoms, 'a' and 'b' are positive constant, $U(r) = \frac{a}{r^5} - \frac{b}{r^3}$. Choose the correct option(s).
- (A) The equilibrium separation between atoms is $r_0 = \sqrt{\frac{5a}{6b}}$.
- (B) The equilibrium separation between atoms is $r_0 = \sqrt{\frac{5a}{3b}}$.
- (C) The oscillation frequency of the molecule if the mass of each atom is m is $\frac{1}{2\pi} \sqrt{\frac{12b}{m} \left(\frac{3b}{5a}\right)^{5/2}}$.
- (D) The oscillation frequency of the molecule if the mass of each atom is m is $\frac{1}{2\pi} \sqrt{\frac{6b}{m} \left(\frac{3b}{5a}\right)^{5/2}}$.
21. Two uniform rods AB and CD, each of length l and mass m are fixed to two gears (Which are discs pivoted to fixed points A and C at same horizontal level) as shown in figure. The system is in vertical plane. When rod AB is slightly rotated by as small angle, rod CD rotates in the opposite sense by another small angle such that there is no slipping and no dissipation of energy at the point of contact of the gears due to their teeth in contact. Neglecting the mass of gears, the time period of small oscillations of the system, is given by $2\pi\sqrt{\frac{20l}{ng}}$. where n is a positive integer and 'g' is acceleration due to gravity. Find n ?



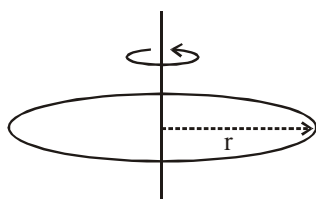
22. A homogeneous solid ball of specific gravity 2 and volume 1 m^3 is suspended on a weightless thread from an end of a homogeneous rod of specific gravity 3, volume 1 m^3 . The rod is placed on the edge of a tumbler with water as shown in figure, so that half of the ball is submerged in water when the system is in equilibrium. Density of water is ρ_0 . Neglecting the surface tension on the boundaries between the ball and water, the ratio y/x of the lengths of the parts of the rod is:



23. A uniform solid sphere of density ρ and radius R starts to move with constant velocity v in a gravity free viscous medium. Viscous drag follows exact stokes law. Due to Viscous drag, heat produced is completely absorbed by the sphere and sphere evaporates uniformly. Latent heat of vaporization of material of sphere is L . Time after radius of the sphere becomes $R/2$ is $\frac{3\rho LR^2}{\beta\eta v^2}$. Here β is an integer. Coefficient of viscosity for the medium is η . Find β .
24. A large tank is filled with water at the rate of $70 \text{ cm}^3/\text{s}$. A hole of cross-section 0.25 cm^2 is punched at the bottom of the tank. Find the maximum height to which the tank can be filled (in cm) (Take $g = 9.8 \text{ m/s}^2$) :
25. A weightless rigid rod with a small iron bob at the end is hinged at point A to the wall so that it can rotate in all directions. The rod is kept in the horizontal position by a vertical inextensible string of length 20 cm , fixed at its mid point. The bob is displaced slightly, perpendicular to the plane of the rod and string. Find period of small oscillations of the system in the form $X \text{ sec.}$ and fill value of X ($g = 10 \text{ m/s}^2$) :



26. A thin spherical soap bubble has surface tension S . It's surface is uniformly charged with charge Q , it's volume is V . It is found that when $Q^2 = 16 n\pi\epsilon_0 SV$, the excess pressure inside the bubble becomes zero. Find n .
27. A ring of radius r made of wire of density ρ is rotated about a stationary vertical axis passing through its centre and perpendicular to the plane of the ring as shown in figure. If the angular velocity (in rad/s) of ring at which the ring breaks is α . Then find out α . The wire breaks at tensile stress σ . Ignore gravity. (Take $\frac{\sigma}{\rho} = 4 \text{ m}^2/\text{s}^2$ and $r = 1 \text{ m}$) :

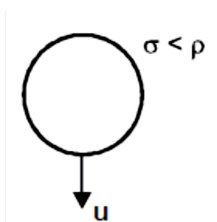


28. Container filled with air under pressure P_0 contains a soap bubble of radius R . The air pressure has been reduced to half isothermally and the new radius of the bubble becomes $\frac{5R}{4}$. If the surface tension of the soap water solution is S . P_0 is found to be $\frac{16nS}{R}$ SI unit. Find n .

29. Composite rod whose upper half has a density $\frac{\rho}{4}$ and lower half has a density of $\frac{3\rho}{2}$ is immersed vertically in a liquid of density ρ . The length of rod which should be immersed so that centre of buoyancy coincides with centre of mass of the rod is $\frac{x\ell}{14}$. Find x .

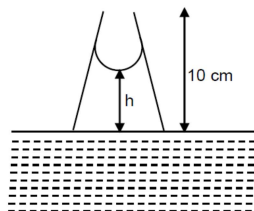


30. A solid sphere made of material of bulk modulus $2 \times 10^8 \text{ N/m}^2$ is surrounded by a liquid in cylindrical container. A massless piston of area 1 m^2 floats on the surface of the liquid. When a mass 6 kg is placed on the piston to compress the liquid then fractional change in the radius of the sphere is given by $x \times 10^{-7}$, then find x .
31. A sphere of density ρ falls vertically downward through a fluid of density σ . At a certain instant its velocity is u . The terminal velocity of the sphere is u_0 . Assuming that Stokes's law for viscous drag is applicable, the instantaneous acceleration of the sphere is found to be $\beta \left(1 - \frac{\sigma}{\rho} \right) \left(1 - \frac{u}{u_0} \right) g$. Here β is an integer. Find β .

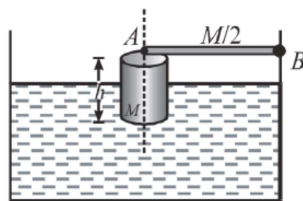


32. A capillary tube with inner cross-section in the form of a square of side 'a' is dipped vertically in a liquid of density ρ and surface tension σ which wets the surface of capillary tube with angle of contact θ . The approximate height to which liquid will be raised in the tube is $\frac{N\sigma \cos \theta}{\rho g}$, where 'N' is: (Neglect the effect of surface tension at the corners of capillary tube) –

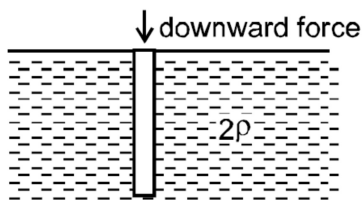
33. Internal diameter of a 10 cm long tapered capillary tube varies uniformly from 1 mm to 0.5 mm from one end to other as shown. The bigger end just touches the surface of a liquid as shown. Surface tension and density of the liquid is $8 \times 10^{-2} \text{ N/m}$ and 1000 kg/m^3 respectively. Angle of contact of the liquid is zero with tube and the meniscus can be assumed hemispherical. Up to what height (in cm) liquid will rise in the tube ? (Take $g = 10 \text{ m/s}^2$)



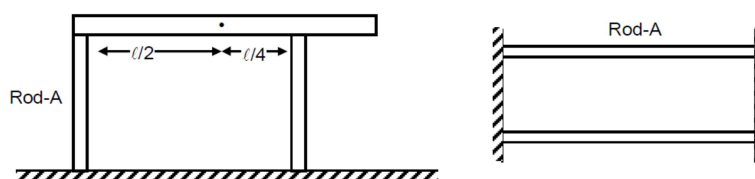
34. A cylindrical block of mass M and height h floats vertically in a liquid which is twice as dense as the block, as shown in the diagram. A uniform rod of mass $M/2$ and length ℓ is hinged to axis of the cylinder at A and to the walls of the vessel at B. The rod can rotate freely about the point B. It is observed that the rod is horizontal in its equilibrium position. If angular frequency of small vertical oscillations of the cylinder is $\sqrt{\frac{12g}{nh}}$. Then n is :



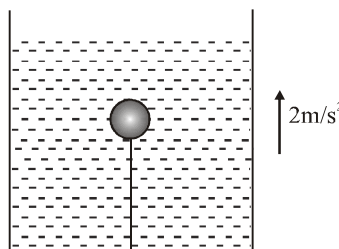
35. A thin uniform elastic rod of natural length L , density ρ and Young's modulus Y is immersed completely in vertical position in a fluid of density 2ρ by applying a vertically downward force at its top end such that the top end of rod coincides with the free surface of fluid. The rod is in equilibrium, then net compression in rod is $\frac{N\rho L^2 g}{4Y}$, where 'N' is: (g is acceleration due to gravity, neglect atmospheric pressure)



36. Two particles A and B are performing SHM along x and y-axis respectively with equal amplitude and frequency of 2 cm & 1 Hz respectively. Equilibrium positions of the particles A and B are at the coordinates (3 cm, 0) and (0, 4 cm) respectively. At $t = 0$, B is at its equilibrium position and moving towards the origin, while A is nearest to the origin and moving away from the origin. Then ratio of minimum and maximum distance between A and B during the motion is 3 : N, where N is :
37. A thin rod of negligible mass and area of cross-section $2 \times 10^{-6} \text{ m}^2$, suspended vertically from one end, has a length of 0.5 m at 200°C . The rod is cooled to 0°C , but prevented from contracting by attaching a mass at the lower end. The value of this mass (in Kg) is : (Young's modulus = 10^{11} N/m^2 , Coefficient of linear expansion 10^{-5} K^{-1} and $g = 10 \text{ m/s}^2$):
38. A uniform heavy plank of length L is resting on two rods of identical length and area of cross section, made up of different materials. It is observed that to keep the planks horizontal, one rod is to be at the end of the plank (say rod A) and another rod is to be at separation $\frac{L}{4}$ from centre of plank as shown in figure. Now the rods are kept between two rigid walls such that separation between the walls is equal to the natural length of the rod. Now temperature of both the rods is gradually increased in same manner, it is observed that thermal stress developed in rod A is always thrice the thermal stress developed in another rod. Ratio of coefficient of linear expansion of rod A to coefficient of linear expansion of another rod will be :



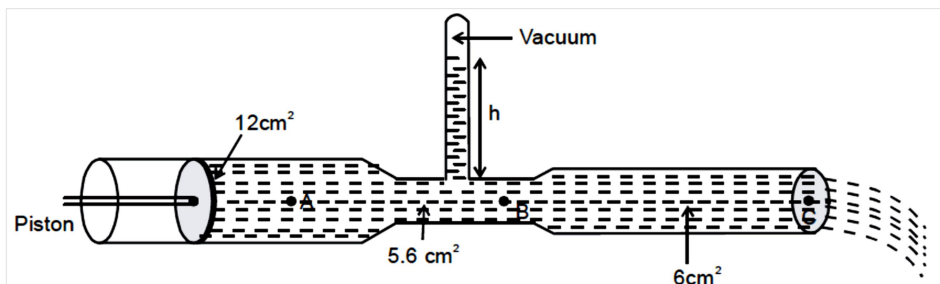
39. A solid sphere of mass $m = 2 \text{ kg}$ and specific gravity $S = 0.5$ is held stationary relative to tank filled with water. The tank is accelerating upward with acceleration 2 m/s^2 . The tension in the string is found to be $(8n) \text{ N}$. Find the value of n. Given density of water = 1000 kg m^{-3} and $g = 10 \text{ m/s}^2$.



40. The radius of soap bubble is R and surface tension of soap solution is T, keeping the temperature constant, the extra energy needed to double the radius of the soap bubble by blowing will be $N\pi R^2 T$, where N is:

Paragraph for Question Nos. 41 to 42

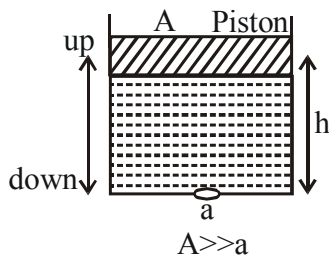
A glass tube has three different cross sectional areas with the values indicated in the figure. A piston at the left end of the tube exerts pressure so that the mercury within the tube flows from the right end with a speed of 8.0 m/s . Three points within the tube are labeled A, B and C. The atmospheric pressure is $1.01 \times 10^5 \text{ N/m}^2$; and the density of mercury is $1.36 \times 10^4 \text{ kg/m}^3$. (use $g = 10 \text{ m/s}^2$) -



41. The pressure at point A is equal to :
 (A) $2.02 \times 10^5 \text{ Pa}$ (B) $2.25 \times 10^5 \text{ Pa}$ (C) $3.26 \times 10^5 \text{ Pa}$ (D) $4.27 \times 10^5 \text{ Pa}$
42. The height h of mercury in the manometer is -
 (A) 136 mm (B) 169 mm (C) 272 mm (D) 366 mm

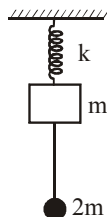
Paragraph for Question Nos. 43 to 44

Artificial satellites go round the earth in their respective orbits. Astronauts are able to float inside the satellite. Inside space of the satellite could be treated as gravity free space. If a beaker filled with a liquid upto height h having a small hole in the bottom is placed in such a satellite, you will wonder the liquid does not come out of the hole. You have to squeeze or force the liquid to come out. Suppose a constant pressure p is applied on the liquid through piston to force it out through the hole. Outside pressure could be neglected in comparison with p . Cross-sectional Area of beaker is too greater than that of hole ($A \gg a$). As the liquid comes out of the hole, it applies a thrust force on the beaker in upward direction.

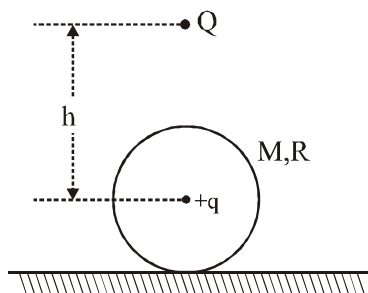


43. Velocity with which liquid comes out of the hole is :
 (A) $\sqrt{\frac{p}{\rho}}$ (B) $\sqrt{\frac{2p}{\rho}}$ (C) $\sqrt{\frac{p}{2\rho}}$ (D) $\frac{A}{a} \sqrt{\frac{p}{2\rho}}$
44. Time taken to empty the beaker is :
 (A) $\frac{Ah}{a} \sqrt{\frac{\rho}{2p}}$ (B) $\frac{Ah}{a} \sqrt{\frac{p}{\rho}}$ (C) $\frac{Ah}{a} \sqrt{\frac{2\rho}{p}}$ (D) $h \sqrt{\frac{p}{2\rho}}$

45. A bob of mass $2m$ hangs by a string attached to the block of mass m of a spring block system. The whole arrangement is in a state of equilibrium. The bob of mass $2m$ is pulled down slowly by a distance x_0 and released.

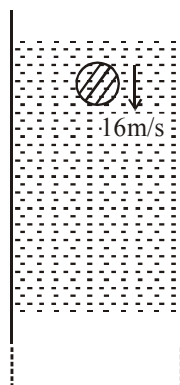


- (A) For $x_0 = \frac{3mg}{k}$ maximum tension in string is $4mg$.
- (B) For $x_0 > \frac{3mg}{k}$, minimum tension in string is mg .
- (C) Frequency of oscillation of system is $\frac{1}{2\pi} \sqrt{\frac{k}{3m}}$, for all non-zero values of x_0 .
- (D) The motion will remain simple harmonic for $x_0 \geq \frac{3mg}{k}$.
46. A point charge having charge $+q$ and of negligible mass is fixed at centre of a uniform solid disc of mass M and radius R which is kept on a fixed sufficiently rough horizontal surface. Another point charge $-Q$ is fixed exactly above centre of disc at height h from the centre of the disc. Whole situation is shown in figure. Now disc is given a small horizontal displacement and released. Time period of SHM of centre of disc is (given that $\frac{Qq}{4\pi\epsilon_0 h^2} < mg$):



- (A) $2\pi h \sqrt{\frac{6\pi\epsilon_0 mh}{Qq}}$ (B) $4\pi h \sqrt{\frac{\pi\epsilon_0 mh}{Qq}}$ (C) $2\pi h \sqrt{\frac{3\pi\epsilon_0 mh}{Qq}}$ (D) $2\pi h \sqrt{\frac{2\pi\epsilon_0 mh}{Qq}}$

47. In a tall viscous liquid column in gravity free space, a solid ball is projected with 16 m/s as shown at $t = 0$. Velocity of the ball reduces to 1 m/s at $t = 4$ s. Choose the incorrect option(s):

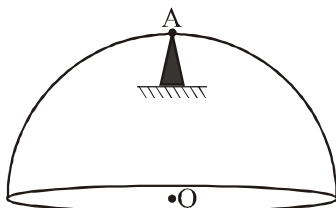


- (A) Velocity of ball at $t = 1$ s is 8 m/s.
 (B) Velocity of ball at $t = 3$ s is 2 m/s.
 (C) If F_1 is average viscous force acting on the ball between $t = 0$ and $t = 2$ s and F_2 is average viscous force on the ball between $t = 2$ and $t = 4$, then value of $\frac{F_1}{F_2}$ is 4.
 (D) If F_1 is average viscous force acting on the ball between $t = 0$ and $t = 2$ s and F_2 is average viscous force on the ball between $t = 2$ and $t = 4$, then value of $\frac{F_1}{F_2}$ is 3.
48. A rod of mass m and length ℓ is acted upon by two forces each of magnitude F as shown in the figure. The length of the rod is ℓ , area of cross section is A , the young's modulus of the material is Y . Select correct alternative :

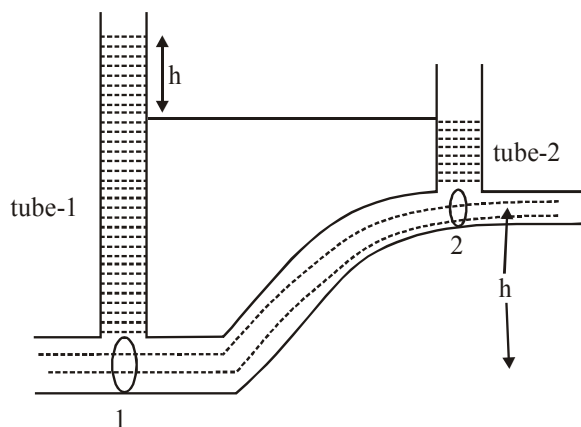


- (A) The stress in rod at mid point of the rod is $\frac{F}{A}$.
 (B) The stress in rod at a distance $\frac{\ell}{4}$ from one end is $\frac{F}{2A}$.
 (C) Net change in length of the rod is $\frac{F\ell}{AY}$.
 (D) Total elastic potential energy stored in the rod is zero.

49. A uniform hemispherical shell of mass M and radius R is made to perform small angular oscillation about point 'A' such that line AO always remains in vertical plane (where O is curvature centre of spherical shell). Frequency of small angular oscillations will be :



- (A) $\frac{1}{2\pi} \sqrt{\frac{6g}{R}}$ (B) $\frac{1}{2\pi} \sqrt{\frac{3g}{R}}$ (C) $\frac{1}{4\pi} \sqrt{\frac{3g}{R}}$ (D) $\frac{1}{2\pi} \sqrt{\frac{g}{R}}$
50. A non - viscous fluid of density ρ is flowing in a tube as shown in figure. Area of section-(1) is double that of section-(2). Centre of mass of section-(2) is h height above the Centre of mass of section-(1) and level of water in tube-1 is ' h ' height above that in tube-2. Then :



- (A) Velocity of fluid of section - (1) is $\sqrt{\frac{gh}{3}}$.
- (B) Velocity of fluid at section - (1) is $\sqrt{\frac{2gh}{3}}$.
- (C) Work done by gravitational force per unit volume from section-(1) to section-(2) is ρgh .
- (D) Work done by elastic forces (pressure) per unit volume from section-(1) to section-(2) is $3\rho gh$.

ANSWER KEY TO DPP # 05 (JEE-ADVANCED)

1.	(A,B,D)	2.	(C,D)	3.	(A,D)	4.	(A,B,C)	5.	(A,D)
6.	(A,B,C)	7.	(B,C)	8.	(A,B,C,D)	9.	(A,D)	10.	(A,B,D)
11.	(A,B,C,D)	12.	(A,B,C,D)	13.	(B,D)	14.	(B,D)	15.	(A,B)
16.	(A,B,C)	17.	(A,B,C,D)	18.	(A,B)	19.	(B,C,D)	20.	(B,C)
21.	18	22.	02	23.	12	24.	40	25.	1.26
26.	6	27.	2	28.	6	29.	9	30.	1
31.	1	32.	4	33.	4	34.	7	35.	6
36.	7	37.	40	38.	6	39.	3	40.	24
41.	(D)	42.	(C)	43.	(B)	44.	(A)	45.	(A)
46.	(A)	47.	(D)	48.	(B)	49.	(C)	50.	(B)